

Optimizing Profitability & Growth

Jaffle Shop Analytics Engineering Hackathon

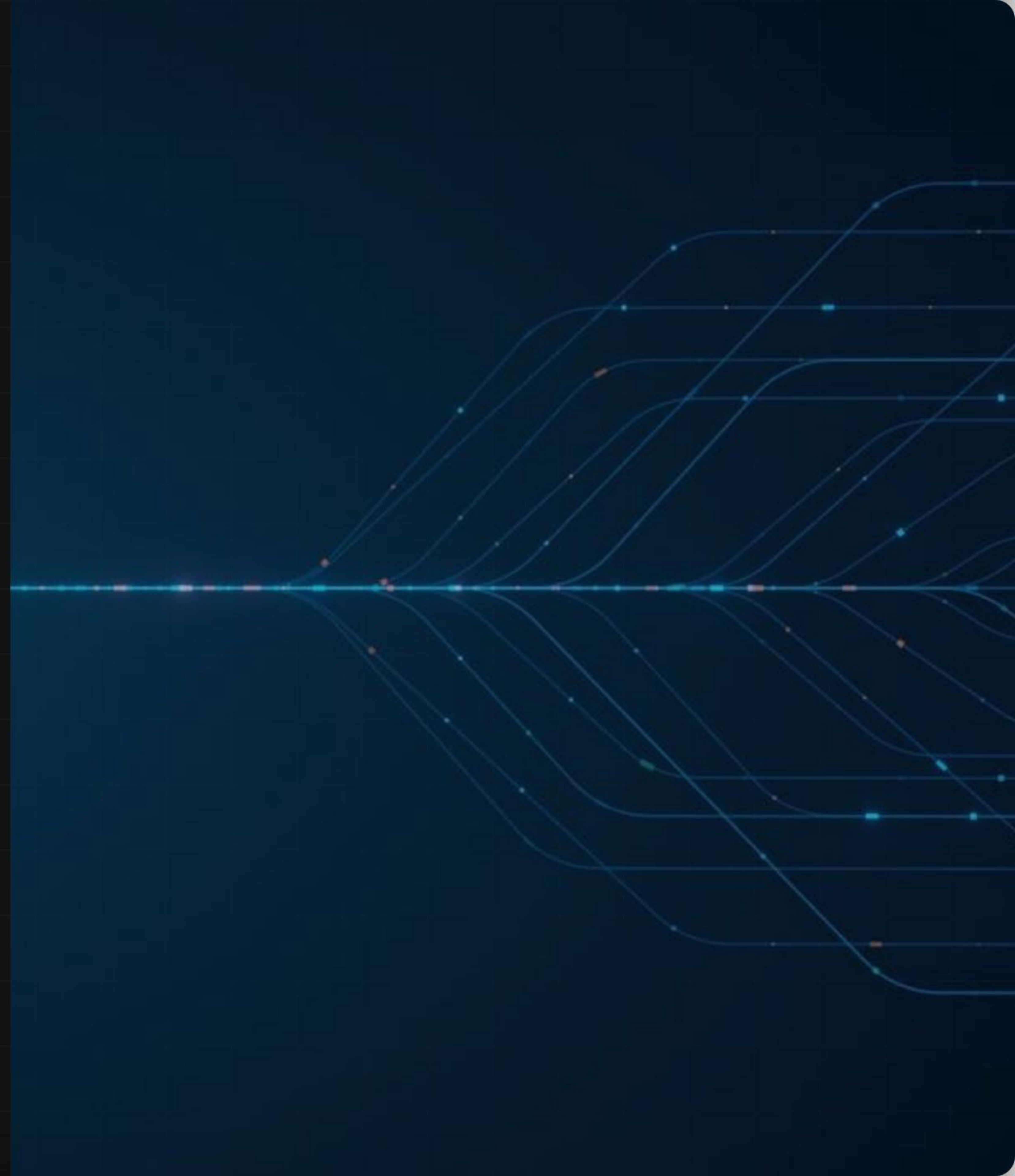
July 10th

The Dual Strategy

Moving beyond raw order volume to uncover true store efficiency and untapped revenue synergies.

Analytics **Architecture**

- **Staging Layer:** Cleaned and mapped raw CSVs for Orders, Items, Products, Supplies, and Stores.
- **Marts Layer:** Developed two highly tested fact tables: `fct_store_margins` and `fct_product_affinity`.
- **Testing rigor:** Enforced unique primary keys, not-null constraints, and strict business logic guardrails.



MarginMaker: **Store Efficiency**

- > **The Question:** Which stores are losing money on high supply costs?
- > **The Engineering:** We joined item-level revenue against localized ingredient/supply costs to find net profit.
- > **Business Logic Test:** We built a custom dbt test to ensure profit margins never register as mathematically impossible negative values.

Profit & Loss Dashboard Infographic



Total Revenue

\$4.8 M ↑ +6.2%

Revenue From Last Month



Total Expenses

\$3.1 M ↑ +2.1%

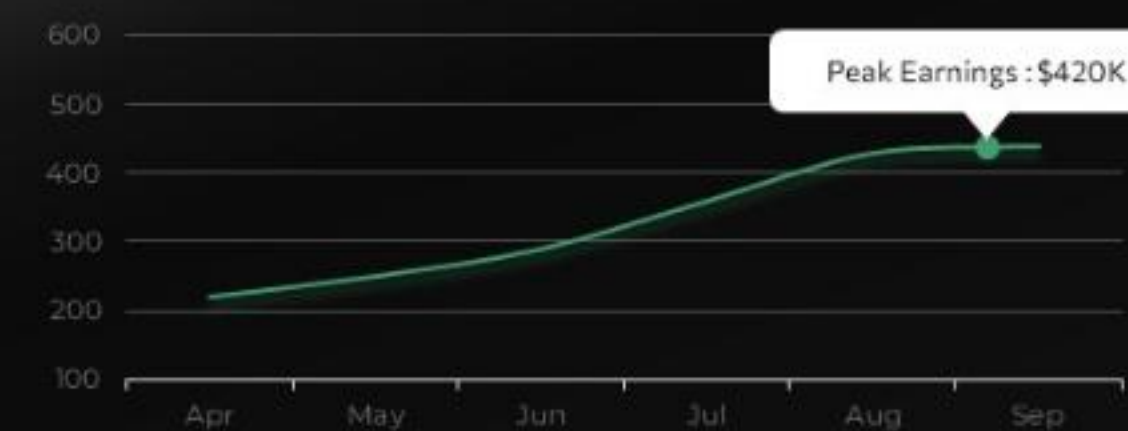
Expenses From Last Month



Net Profit

\$1.7 M ↑ +35.4%

Profit Margin From Last Month



Top Revenue Sources

Software Sales - \$2.6M

+8.4%

Services - \$720K

+3.6%

Subscriptions - \$1.1M

+5.1%

Other Income - \$380K

-2.2%

Expense Ratio



■ Salaries
■ Operations
■ Marketing
■ Admin
■ Taxes

The Insight & Action

+18%

PROFIT VARIANCE

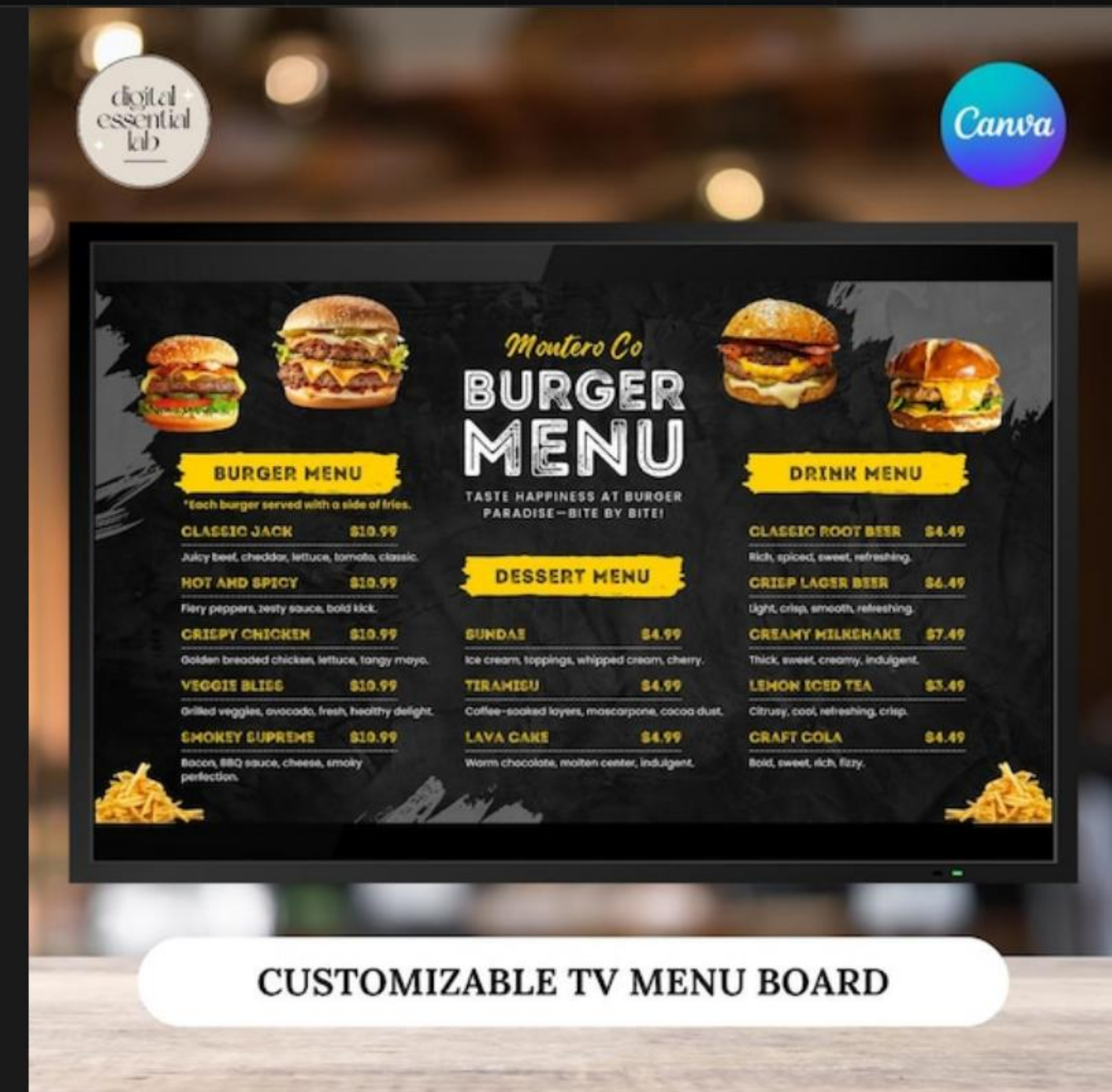
Efficiency over Volume

Insight: Store A processes fewer total orders than Store B, yet maintains an 18% higher profit margin due to superior supply-cost controls.

Actionable Next Step: Execute an immediate audit of Store B's local inventory waste and mandate a transition to Store A's supplier network.

BasketWeaver: Cross-Selling

- > **The Question:** Which products organically drive the sales of others?
- > **The Engineering:** We utilized a complex self-joining CTE to execute a market basket analysis across all historical orders.
- > **Business Logic Test:** We tested that `'product_a' != 'product_b'` to guarantee our pipeline never artificially pairs a product with itself.



The Insight & Action

35%

CO-OCCURRENCE RATE

Unlocking Free Revenue

Insight: Product X and Product Y are naturally purchased together in 35% of all orders where either item is present.

Actionable Next Step: Redesign the digital menu and POS system to offer a '1-Click Combo' upsell, instantly increasing Average Order Value (AOV).

Questions?

Ready to deploy these models to production.

 `github.com/your-username/jaffle-shop-dbt`

Image Sources



https://static.vecteezy.com/system/resources/thumbnails/081/299/177/small_2x/abstract-data-flow-branching-lines-on-dark-blue-background-with-glowing-cyan-green-and-orange-data-particles-technology-concept-for-ai-pipeline-bigdata-streaming-neural-network-and-digital-video.jpg

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